# Math 4441: Probability and Statistics

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# Chapter 3: Well-Known Random Variables and Distributions
# (Part 2 – Continuous Well-known RVs)
# Lectures 17 – 18

**Math 4441: Probability and Statistics**

References:


1. Goodman and Yates – Introduction to Probability and Stochastic Processes, 3rd Edition
2. Scott Miller and Donald Childers – Probability and Random Processes, 2nd Edition
3. Bertsekas and Tsitsiklis – Introduction to Probability, 2nd Edition

Well-Known Continuous Random Variables

1. Uniform Continuous Random Variables
2. Exponential Random Variables
3. Gamma Random Variables
4. Beta Random Variables
5. Normal Random Variables
6. Log Normal Random Variables
7. Laplace Random Variables

Uniform Continuous Random Variables

- Consider a finite interval of length $[a, b]$
- Assume the interval is divided into $n$ equal length sub-intervals

- Suppose you pick a value randomly from the interval
  - Random variable $X$ denotes the value
- $X$ is said to be uniformly distributed, if the probability that $X$ takes value from any of the subintervals is equal
  - Probability per unit length for all subintervals is same
  - Probability density is constant

Definition: (*Uniform Continuous Random Variables*) A random variable $X$ is said to be uniformly distributed for an interval $[a, b]$, if the probability density is a constant for the entire interval. The probability density function of $X$ is

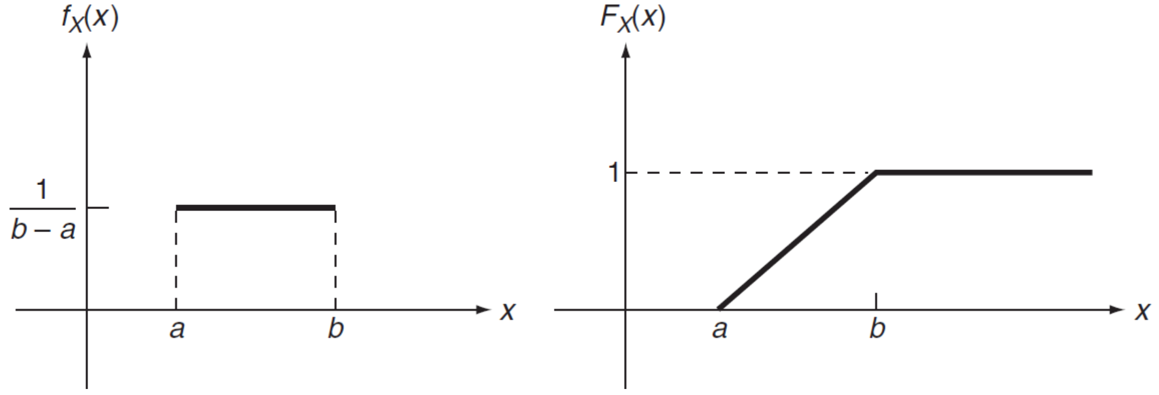
$$f_X(x) = \begin{cases} C, & \text{for } a \leq X \leq b; \\ 0, & \text{otherwise.} \end{cases}$$

$$\int_{-\infty}^{+\infty} f_X(x)dx = 1$$
$$\Rightarrow \int_a^b C dx = 1$$
$$\Rightarrow cx\Big|_a^b = 1$$
$$\Rightarrow c(b-a) = 1$$
$$\Rightarrow c = \frac{1}{b-a}.$$

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & \text{for } a \leq X \leq b; \\ 0, & \text{otherwise.} \end{cases}$$

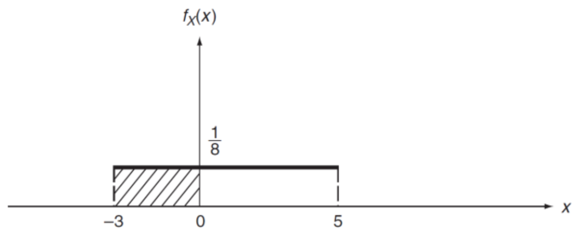
$$
\begin{aligned}
F_X(x) &= \int_a^x f_X(x)dx \\
&= \int_a^x \frac{1}{b-a}dx \\
&= \frac{x}{b-a}\Big|_a^x \\
&= \frac{x-a}{b-a}.
\end{aligned}
$$

$$
F_X(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & \text{for } a \leq X \leq b \\ 1, & x \geq 1. \end{cases}
$$



$f_X(x)$
$\frac{1}{b-a}$
a
b
x
$F_X(x)$
1
a
b
x

Example: Owing to unpredictable traffic situations, the time required by a certain student to travel from her home to her morning class is uniformly distributed between 22 and 30 minutes. If she leaves home at precisely 7.35 a.m., what is the probability that she will not be late for class, which begins promptly at 8:00 a.m.?




$$f_X(x) = \begin{cases} \frac{1}{8}, & \text{for } -3 \leq x \leq 5; \\ 0, & otherwise. \end{cases}$$

$P[-3 \leq X \leq 0] = 3 \times \frac{1}{8} = \frac{3}{8}$

If a random variable $X$ is uniformly distributed over an interval $A$, then the probability of $X$ taking values in a subinterval $B$ is

$$P[X \in B] = \frac{length\ of\ B}{length\ of\ A}$$

**Exponential Random Variables**

- Exponential distribution is related to the Poisson distribution
- A Poisson random variable represents
  - probability of the # of arrivals in a certain period of time
  - given average arrival rate of that time
- For such experiments, the number of arrivals is not the only random quantity, there are other random quantities of interest
  - Time to the first arrival
  - Time between successive intervals
    - Interarrival times

- Customers arrive in a shopping mall following the Poisson distribution
    - at a rate $\lambda$ customers per hour
    - $n$ customers arrive by time $t$
- Let $X$ denote the time elapsed before the arrival of the first customer
- $X_i$ represent the time between the arrivals of the $i$-th and $(i-1)$-th customers
- $X_1, X_2, \dots$ are random values and are called interarrival times
    - And are continuous random variables

PDF and CDF of First arrival time, say $X$

CDF of the random variable $X$,

$$F_X(x) = P[X \leq x]$$

Complementary CDF of $X$, $\bar{F}_X(x) = 1 - F_X(x) = P[X \geq x]$

$$\bar{F}_X(t) = P[X \geq t]$$
$$= P[0 \text{ arrival in the interval } (0, t)]$$

$$P[N = n] = \frac{e^{-\lambda t}(\lambda t)^n}{n!}, n = 0, 1, 2, \ldots$$

$$\bar{F}_X(x) = \frac{e^{-\lambda t}(\lambda t)^n}{n!}$$
$$= e^{-\lambda x}.$$

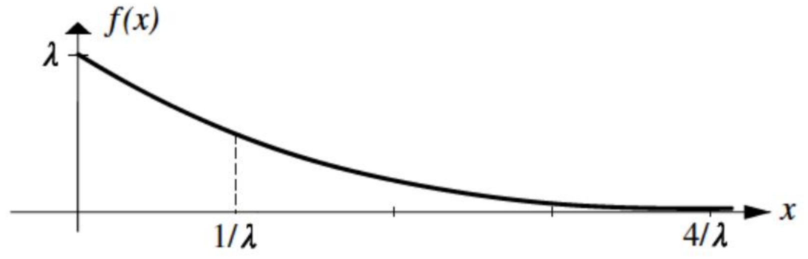
$$F_X(x) = 1 - e^{-\lambda t}, t \geq 0$$

$$f_X(t) = 0 - e^{-\lambda t}(-\lambda) = \lambda e^{\lambda t}, t \geq 0$$

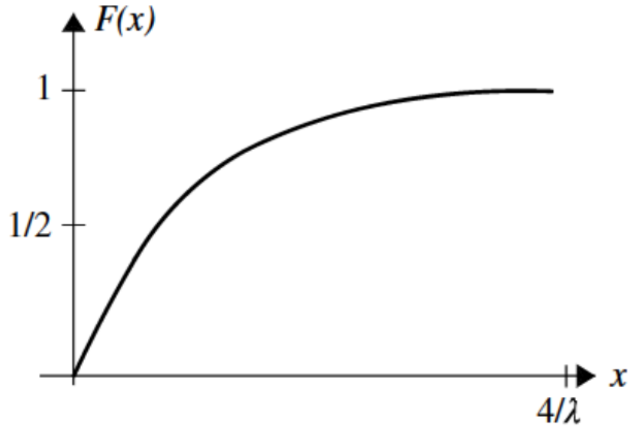
$$E(X) = \Big[-xe^{-\lambda x}\Big]_0^\infty + \int_0^\infty e^{-\lambda x}\,dx = 0 - \Big[\frac{1}{\lambda}e^{-\lambda x}\Big]_0^\infty = \frac{1}{\lambda}.$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx = \int_0^\infty x^2(\lambda e^{-\lambda x})\,dx = \frac{2}{\lambda^2}$$

$$\mathrm{Var}(X) = E(X^2) - \big[E(X)\big]^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$



f(x)
λ
1/λ
4/λ
x

F(x)
1
1/2
4/λ
x

Example: suppose that every three months, on average, an earthquake occurs in a city. What is the probability that the next earthquake occurs after three but before seven months?

$$F(t) = P(X \leq t) = 1 - e^{-t/3} \quad \text{for } t > 0,$$

$$P(3 < X < 7) = F(7) - F(3) = (1 - e^{-7/3}) - (1 - e^{-1}) \approx 0.27.$$

Relation between Poisson and Exponential

$X \sim \text{Poisson}\ (\lambda)$

$X \sim \text{Exp}\ (\lambda)$

Same Parameter $\lambda$

Geometric vs Exponential: Pascal vs ?

$n$-Erlang and Gamma Random Variables

- Gamma distribution is related to Poisson distribution as well
- Exponential distribution models the length of the time of the first arrival.
    - Also, the length of the time of successive arrivals
- Gamma distribution is the generalization of the exponential distribution
    - models the length of the time required to get the $n$-th arrival

Let $X_i, i = 1, 2, 3, \ldots, n$, denote the interarrival times of the events of the Poisson distribution, where $X_1$ be the time required for the arrival of the first event, and for $i \geq 2$, $X_i$ be the time between the arrivals of the $i$-th and the $(i-1)$-th events. Let $X$ represent the time required for the occurrence of the $n$-th event.

$$X = X_1 + X_2 + \cdots + X_n.$$

$X$ is said to have the gamma distribution with parameters $n$ and $\lambda$.

$X \sim$ gamma $(n, \lambda)$, if $X_i$s are IID exponential

To derive the CDF of gamma distribution

- The event $\{X \leq t\}$ occurs, iif the $n$-th event occurs in $(0, t)$
- # of events, $n$, occurs in $(0, t)$ is at least $n$
- $n$ or more Poisson arrivals occur in the interval $(0, t)$

$$
\begin{aligned}
F_X(t) = P[X \leq t] = P[N \geq n] \\
= \sum_{i=n}^{\infty} \frac{e^{-\lambda t}(\lambda t)^i}{i!}.
\end{aligned}
$$

$$
\begin{aligned}
f_X(t) &= \frac{d}{dt}F_X(t) \\
&= \sum_{i=n}^{\infty}\left[-\lambda\frac{e^{-\lambda t}(\lambda t)^i}{i!} + \lambda\frac{e^{-\lambda t}(\lambda t)^{i-1}}{(i-1)!}\right] \\
&= -\sum_{i=n}^{\infty}\lambda\frac{e^{-\lambda t}(\lambda t)^i}{i!} + \lambda\frac{e^{-\lambda t}(\lambda t)^{n-1}}{(n-1)!} + \sum_{i=n+1}^{\infty}\lambda\frac{e^{-\lambda t}(\lambda t)^{i-1}}{(i-1)!} \\
&= -\sum_{i=n}^{\infty}\lambda\frac{e^{-\lambda t}(\lambda t)^i}{i!} + \lambda\frac{e^{-\lambda t}(\lambda t)^{n-1}}{(n-1)!} + \sum_{i=n}^{\infty}\lambda\frac{e^{-\lambda t}(\lambda t)^{i}}{(i)!} \\
&= \lambda\frac{e^{-\lambda t}(\lambda t)^{n-1}}{(n-1)!}.
\end{aligned}
$$

**n-Erlang Distribution** with parameters $n$ and $\lambda$

Works only for positive integer values of n

The probability density function

$$f(x) = \begin{cases} \lambda e^{-\lambda x} \dfrac{(\lambda x)^{n-1}}{(n-1)!} & \text{if } x \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

is called the gamma (or n-Erlang) distribution with parameter $(n, \lambda)$

Gamma Function: For each positive real number, $\alpha$, i.e., $\alpha > 0$, the function $\Gamma: (0, \alpha) \rightarrow R$, defined by

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$$

Is known as the gamma function

For $\alpha > 1$

$$\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1).$$

$\Gamma(1) = 1$

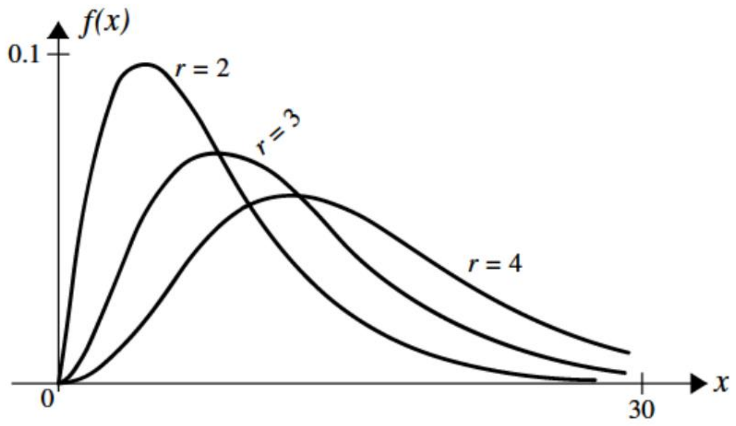
$\Gamma(n) = (n-1)!$

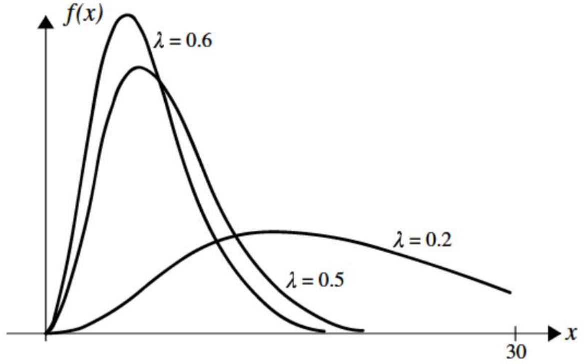
$$
\begin{aligned}
\Gamma(2) &= (2-1)\Gamma(2-1) = 1.\Gamma(1) = 1 = 1! \\
\Gamma(3) &= (3-1)\Gamma(3-1) = 2.\Gamma(1) = 2.1! = 2! \\
\Gamma(4) &= (4-1)\Gamma(4-1) = 3.\Gamma(3) = 3.2! = 3! \\
&\vdots \\
\Gamma(n) &= (n-1)\Gamma(n-1) = (n-1)(n-2)\Gamma(n-3) \\
&= (n-1)(n-2)\ldots 1\Gamma(1) \\
&= (n-1)!
\end{aligned}
$$

*A random variable X with probability density function*

$$f(x) = \begin{cases} \dfrac{\lambda e^{-\lambda x}(\lambda x)^{r-1}}{\Gamma(r)} & \text{if } x \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

*is said to have a **gamma** distribution with parameters* $(r, \lambda)$, $\lambda > 0$, $r > 0$


f(x)
0.1
r = 2
r = 3
r = 4
0
30
x


f(x)
λ = 0.6
λ = 0.2
λ = 0.5
30
x

## Expected Value and Variance

$$E(X) = \frac{r}{\lambda}, \qquad \mathrm{Var}(X) = \frac{r}{\lambda^2}, \qquad \sigma_X = \frac{\sqrt{r}}{\lambda}$$

Example: Customers arrive at a restaurant at a Poisson rate of 12 per hour. If the restaurant makes a profit only after 30 customers have arrived, what is the expected length of time until the restaurant starts to make a profit?

$X \rightarrow$ time until the restaurant starts to make profit

$\lambda = 12/hour, \;\; r = 31$

$E[X] = \frac{r}{\lambda} = \frac{31}{12} = 2\ hours\ 35\ minutes$

Beta Distribution: A random variable $X$ with parameters $\alpha \geq 0$ and $\beta \geq 0$ is called a beta random variable if its PDF is given by

and $X$ has the beta distribution with parameter $\alpha$ and $\beta$ with the following terms:

1. the distribution has the name beta.
2. the distribution has a parameter named beta ($\beta$).
3. the distribution has a function named beta given by

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

A random variable $X$ is said to be beta with parameters $(\alpha, \beta), \alpha > 0, \beta > 0$ if its PDF is given by

$$f_X(x) = f(x) = \begin{cases} \dfrac{1}{B(\alpha,\beta)} x^{\overset{\alpha-1}{\cancel{1-\alpha}}}(1-x)^{\overset{\beta-1}{\cancel{1-\beta}}}, & \cancel{x < 0} \quad 0 < x < 1 \\ 0, & \cancel{x \geq 0} \quad o/t \end{cases}$$

Where

$$B(\alpha,\beta) = \int_0^1 x^{\overset{\alpha-1}{\cancel{1-\alpha}}}(1-x)^{\overset{\beta-1}{\cancel{1-\beta}}}dx$$

Expected value and variance of a beta random variable

$$E[X] = \frac{\alpha}{\alpha + \beta}$$

$$Var[X] = \frac{\alpha\beta}{(\alpha + \beta)^2 \ast (\alpha + \beta + 1)}$$

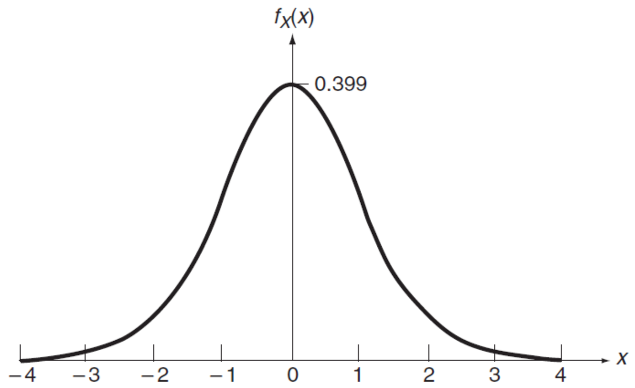
Gaussian or Normal Random Variables: A random variable $X$ is *Gaussian* or *Normal* if its PDF $f_X(x)$ is of the form

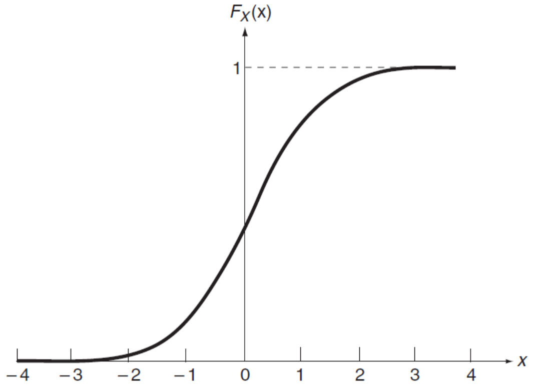
$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \qquad \text{for } -\infty < x < \infty$$

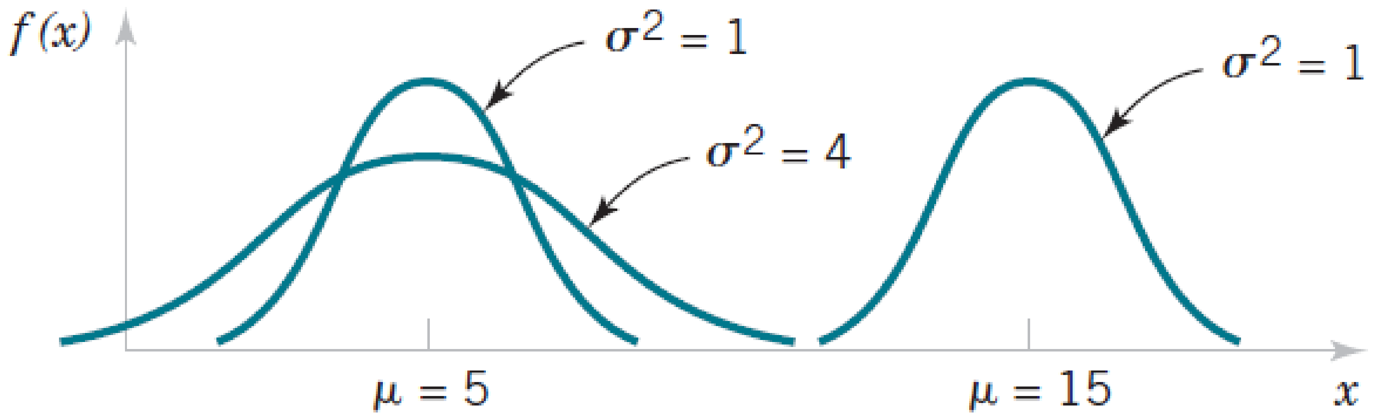
where $\mu$ and $\sigma$ are two parameters, with $\sigma > 0$

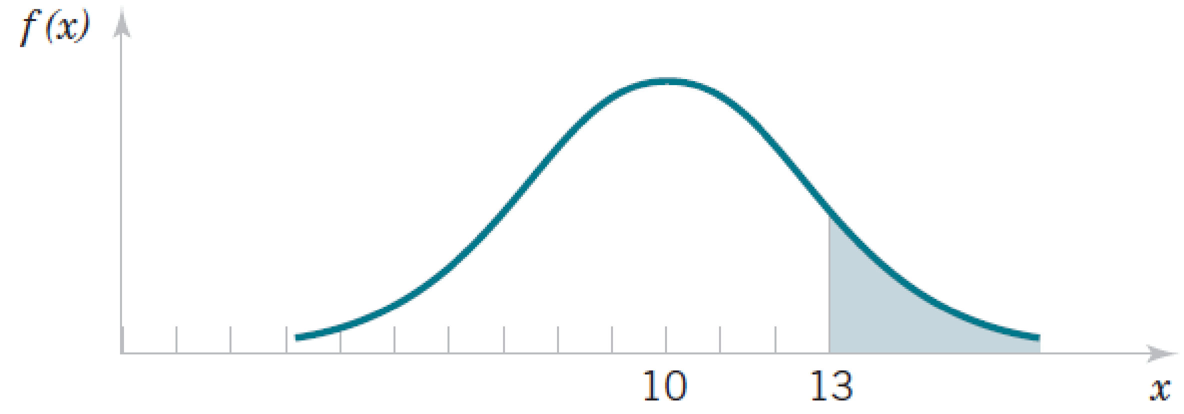
The cumulative distribution function of Gaussian random variable is given by

$$F_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{x} \exp\left[-\frac{(u-\mu)^2}{2\sigma^2}\right] du, \text{for } -\infty < x < \infty.$$

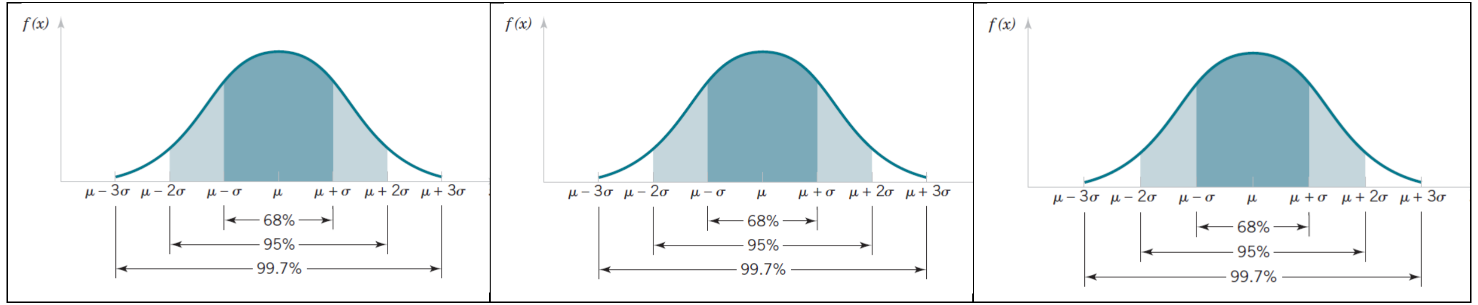

$f_X(x)$
0.399
−4
−3
−2
−1
0
1
2
3
4
x


$F_X(x)$
1
−4
−3
−2
−1
0
1
2
3
4
x


f (x)
σ2 = 1
σ2 = 4
σ2 = 1
μ = 5
μ = 15
x




Probability that $X > 13$ for normal random variable with $\mu = 10$ and $\sigma = 4$




Probability associated with a normal random variable: Area under the curve

$$P(\mu - \sigma < X < \mu + \sigma) = 0.6827$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.9545$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.9973$$

A closed form solution for the CDF of the normal random variable does not exist, and hence, the probability associated with a normal random variable cannot be calculated.

A Table is usually used to get the value. However, the table only enlists the CDF for a special Normal random variable, where $\mu = 0$ and $\sigma^2 = 1$

A normal random variable with zero mean and unit variance is known as the standard normal random variable.

Standard Normal Random Variable: A normal random variable is said to be standard normal random variable, if its expected/mean value is zero and variance is 1, and therefore, it has the following pdf

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}},\ -\infty$$

The CDF of standard normal random variable, $\Phi(z)$, defined as

$$\Phi(z) = P[Z \leq z]$$

The complementary CDF of standard normal random variable, $Q(z)$, is defined as

$$Q(z) = P[Z > z]$$

Symmetric behavior of the standard normal (or normal) curve

$$\Phi(-z) = 1 - \Phi(z)$$

$$\Phi^{-1}(p) = 1 - \Phi^{-1}(p)$$

$$\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2} du$$

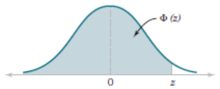



**TABLE III** Cumulative Standard Normal Distribution (*continued*)

| $z$ | 0.00 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.06 | 0.07 | 0.08 | 0.09 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | 0.500000 | 0.503989 | 0.507978 | 0.511967 | 0.515953 | 0.519939 | 0.532922 | 0.527903 | 0.531881 | 0.535856 |
| 0.1 | 0.539828 | 0.543795 | 0.547758 | 0.551717 | 0.555760 | 0.559618 | 0.563559 | 0.567495 | 0.571424 | 0.575345 |
| 0.2 | 0.579260 | 0.583166 | 0.587064 | 0.590954 | 0.594835 | 0.598706 | 0.602568 | 0.606420 | 0.610261 | 0.614092 |
| 0.3 | 0.617911 | 0.621719 | 0.625516 | 0.629300 | 0.633072 | 0.636831 | 0.640576 | 0.644309 | 0.648027 | 0.651732 |
| 0.4 | 0.655422 | 0.659097 | 0.662757 | 0.666402 | 0.670031 | 0.673645 | 0.677242 | 0.680822 | 0.684386 | 0.687933 |
| 0.5 | 0.691462 | 0.694974 | 0.698468 | 0.701944 | 0.705401 | 0.708840 | 0.712260 | 0.715661 | 0.719043 | 0.722405 |
| 0.6 | 0.725747 | 0.729069 | 0.732371 | 0.735653 | 0.738914 | 0.742154 | 0.745373 | 0.748571 | 0.751748 | 0.754903 |
| 0.7 | 0.758036 | 0.761148 | 0.764238 | 0.767305 | 0.770350 | 0.773373 | 0.776373 | 0.779350 | 0.782305 | 0.785236 |
| 0.8 | 0.788145 | 0.791030 | 0.793892 | 0.796731 | 0.799546 | 0.802338 | 0.805106 | 0.807850 | 0.810570 | 0.813267 |
| 0.9 | 0.815940 | 0.818589 | 0.821214 | 0.823815 | 0.826391 | 0.828944 | 0.831472 | 0.833977 | 0.836457 | 0.838913 |
| 1.0 | 0.841345 | 0.843752 | 0.846136 | 0.848495 | 0.850830 | 0.853141 | 0.855428 | 0.857690 | 0.859929 | 0.862143 |
| 1.1 | 0.864334 | 0.866500 | 0.868643 | 0.870762 | 0.872857 | 0.874928 | 0.876976 | 0.878999 | 0.881000 | 0.882977 |
| 1.2 | 0.884930 | 0.886860 | 0.888767 | 0.890651 | 0.892512 | 0.894350 | 0.896165 | 0.897958 | 0.899727 | 0.901475 |
| 1.3 | 0.903199 | 0.904902 | 0.906582 | 0.908241 | 0.909877 | 0.911492 | 0.913085 | 0.914657 | 0.916207 | 0.917736 |
| 1.4 | 0.919243 | 0.920730 | 0.922196 | 0.923641 | 0.925066 | 0.926471 | 0.927855 | 0.929219 | 0.930563 | 0.931888 |
| 1.5 | 0.933193 | 0.934478 | 0.935744 | 0.936992 | 0.938220 | 0.939429 | 0.940620 | 0.941792 | 0.942947 | 0.944083 |
| 1.6 | 0.945201 | 0.946301 | 0.947384 | 0.948449 | 0.949497 | 0.950529 | 0.951543 | 0.952540 | 0.953521 | 0.954486 |
| 1.7 | 0.955435 | 0.956367 | 0.957284 | 0.958185 | 0.959071 | 0.959941 | 0.960796 | 0.961636 | 0.962462 | 0.963273 |
| 1.8 | 0.964070 | 0.964852 | 0.965621 | 0.966375 | 0.967116 | 0.967843 | 0.968557 | 0.969258 | 0.969946 | 0.970621 |
| 1.9 | 0.971283 | 0.971933 | 0.972571 | 0.973197 | 0.973810 | 0.974412 | 0.975002 | 0.975581 | 0.976148 | 0.976705 |
| 2.0 | 0.977250 | 0.977784 | 0.978308 | 0.978822 | 0.979325 | 0.979818 | 0.980301 | 0.980774 | 0.981237 | 0.981691 |
| 2.1 | 0.982136 | 0.982571 | 0.982997 | 0.983414 | 0.983823 | 0.984222 | 0.984614 | 0.984997 | 0.985371 | 0.985738 |
| 2.2 | 0.986097 | 0.986447 | 0.986791 | 0.987126 | 0.987455 | 0.987776 | 0.988089 | 0.988396 | 0.988696 | 0.988989 |
| 2.3 | 0.989276 | 0.989556 | 0.989830 | 0.990097 | 0.990358 | 0.990613 | 0.990863 | 0.991106 | 0.991344 | 0.991576 |
| 2.4 | 0.991802 | 0.992024 | 0.992240 | 0.992451 | 0.992656 | 0.992857 | 0.993053 | 0.993244 | 0.993431 | 0.993613 |
| 2.5 | 0.993790 | 0.993963 | 0.994132 | 0.994297 | 0.994457 | 0.994614 | 0.994766 | 0.994915 | 0.995060 | 0.995201 |
| 2.6 | 0.995339 | 0.995473 | 0.995604 | 0.995731 | 0.995855 | 0.995975 | 0.996093 | 0.996207 | 0.996319 | 0.996427 |
| 2.7 | 0.996533 | 0.996636 | 0.996736 | 0.996833 | 0.996928 | 0.997020 | 0.997110 | 0.997197 | 0.997282 | 0.997365 |
| 2.8 | 0.997445 | 0.997523 | 0.997599 | 0.997673 | 0.997744 | 0.997814 | 0.997882 | 0.997948 | 0.998012 | 0.998074 |
| 2.9 | 0.998134 | 0.998193 | 0.998250 | 0.998305 | 0.998359 | 0.998411 | 0.998462 | 0.998511 | 0.998559 | 0.998605 |
| 3.0 | 0.998650 | 0.998694 | 0.998736 | 0.998777 | 0.998817 | 0.998856 | 0.998893 | 0.998930 | 0.998965 | 0.998999 |
| 3.1 | 0.999032 | 0.999065 | 0.999096 | 0.999126 | 0.999155 | 0.999184 | 0.999211 | 0.999238 | 0.999264 | 0.999289 |
| 3.2 | 0.999313 | 0.999336 | 0.999359 | 0.999381 | 0.999402 | 0.999423 | 0.999443 | 0.999462 | 0.999481 | 0.999499 |
| 3.3 | 0.999517 | 0.999533 | 0.999550 | 0.999566 | 0.999581 | 0.999596 | 0.999610 | 0.999624 | 0.999638 | 0.999650 |
| 3.4 | 0.999663 | 0.999675 | 0.999687 | 0.999698 | 0.999709 | 0.999720 | 0.999730 | 0.999740 | 0.999749 | 0.999758 |
| 3.5 | 0.999767 | 0.999776 | 0.999784 | 0.999792 | 0.999800 | 0.999807 | 0.999815 | 0.999821 | 0.999828 | 0.999835 |
| 3.6 | 0.999841 | 0.999847 | 0.999853 | 0.999858 | 0.999864 | 0.999869 | 0.999874 | 0.999879 | 0.999883 | 0.999888 |
| 3.7 | 0.999892 | 0.999896 | 0.999900 | 0.999904 | 0.999908 | 0.999912 | 0.999915 | 0.999918 | 0.999922 | 0.999925 |
| 3.8 | 0.999928 | 0.999931 | 0.999933 | 0.999936 | 0.999938 | 0.999941 | 0.999943 | 0.999946 | 0.999948 | 0.999950 |
| 3.9 | 0.999952 | 0.999954 | 0.999956 | 0.999958 | 0.999959 | 0.999961 | 0.999963 | 0.999964 | 0.999966 | 0.999967 |

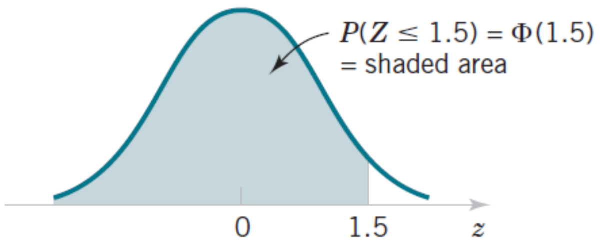
| $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.00 | 0.5000 | 0.50 | 0.6915 | 1.00 | 0.8413 | 1.50 | 0.9332 | 2.00 | 0.97725 | 2.50 | 0.99379 |
| 0.01 | 0.5040 | 0.51 | 0.6950 | 1.01 | 0.8438 | 1.51 | 0.9345 | 2.01 | 0.97778 | 2.51 | 0.99396 |
| 0.02 | 0.5080 | 0.52 | 0.6985 | 1.02 | 0.8461 | 1.52 | 0.9357 | 2.02 | 0.97831 | 2.52 | 0.99413 |
| 0.03 | 0.5120 | 0.53 | 0.7019 | 1.03 | 0.8485 | 1.53 | 0.9370 | 2.03 | 0.97882 | 2.53 | 0.99430 |
| 0.04 | 0.5160 | 0.54 | 0.7054 | 1.04 | 0.8508 | 1.54 | 0.9382 | 2.04 | 0.97932 | 2.54 | 0.99446 |
| 0.05 | 0.5199 | 0.55 | 0.7088 | 1.05 | 0.8531 | 1.55 | 0.9394 | 2.05 | 0.97982 | 2.55 | 0.99461 |
| 0.06 | 0.5239 | 0.56 | 0.7123 | 1.06 | 0.8554 | 1.56 | 0.9406 | 2.06 | 0.98030 | 2.56 | 0.99477 |
| 0.07 | 0.5279 | 0.57 | 0.7157 | 1.07 | 0.8577 | 1.57 | 0.9418 | 2.07 | 0.98077 | 2.57 | 0.99492 |
| 0.08 | 0.5319 | 0.58 | 0.7190 | 1.08 | 0.8599 | 1.58 | 0.9429 | 2.08 | 0.98124 | 2.58 | 0.99506 |
| 0.09 | 0.5359 | 0.59 | 0.7224 | 1.09 | 0.8621 | 1.59 | 0.9441 | 2.09 | 0.98169 | 2.59 | 0.99520 |
| 0.10 | 0.5398 | 0.60 | 0.7257 | 1.10 | 0.8643 | 1.60 | 0.9452 | 2.10 | 0.98214 | 2.60 | 0.99534 |
| 0.11 | 0.5438 | 0.61 | 0.7291 | 1.11 | 0.8665 | 1.61 | 0.9463 | 2.11 | 0.98257 | 2.61 | 0.99547 |
| 0.12 | 0.5478 | 0.62 | 0.7324 | 1.12 | 0.8686 | 1.62 | 0.9474 | 2.12 | 0.98300 | 2.62 | 0.99560 |
| 0.13 | 0.5517 | 0.63 | 0.7357 | 1.13 | 0.8708 | 1.63 | 0.9484 | 2.13 | 0.98341 | 2.63 | 0.99573 |
| 0.14 | 0.5557 | 0.64 | 0.7389 | 1.14 | 0.8729 | 1.64 | 0.9495 | 2.14 | 0.98382 | 2.64 | 0.99585 |
| 0.15 | 0.5596 | 0.65 | 0.7422 | 1.15 | 0.8749 | 1.65 | 0.9505 | 2.15 | 0.98422 | 2.65 | 0.99598 |
| 0.16 | 0.5636 | 0.66 | 0.7454 | 1.16 | 0.8770 | 1.66 | 0.9515 | 2.16 | 0.98461 | 2.66 | 0.99609 |
| 0.17 | 0.5675 | 0.67 | 0.7486 | 1.17 | 0.8790 | 1.67 | 0.9525 | 2.17 | 0.98500 | 2.67 | 0.99621 |
| 0.18 | 0.5714 | 0.68 | 0.7517 | 1.18 | 0.8810 | 1.68 | 0.9535 | 2.18 | 0.98537 | 2.68 | 0.99632 |
| 0.19 | 0.5753 | 0.69 | 0.7549 | 1.19 | 0.8830 | 1.69 | 0.9545 | 2.19 | 0.98574 | 2.69 | 0.99643 |
| 0.20 | 0.5793 | 0.70 | 0.7580 | 1.20 | 0.8849 | 1.70 | 0.9554 | 2.20 | 0.98610 | 2.70 | 0.99653 |
| 0.21 | 0.5832 | 0.71 | 0.7611 | 1.21 | 0.8869 | 1.71 | 0.9564 | 2.21 | 0.98645 | 2.71 | 0.99664 |
| 0.22 | 0.5871 | 0.72 | 0.7642 | 1.22 | 0.8888 | 1.72 | 0.9573 | 2.22 | 0.98679 | 2.72 | 0.99674 |
| 0.23 | 0.5910 | 0.73 | 0.7673 | 1.23 | 0.8907 | 1.73 | 0.9582 | 2.23 | 0.98713 | 2.73 | 0.99683 |
| 0.24 | 0.5948 | 0.74 | 0.7704 | 1.24 | 0.8925 | 1.74 | 0.9591 | 2.24 | 0.98745 | 2.74 | 0.99693 |
| 0.25 | 0.5987 | 0.75 | 0.7734 | 1.25 | 0.8944 | 1.75 | 0.9599 | 2.25 | 0.98778 | 2.75 | 0.99702 |
| 0.26 | 0.6026 | 0.76 | 0.7764 | 1.26 | 0.8962 | 1.76 | 0.9608 | 2.26 | 0.98809 | 2.76 | 0.99711 |
| 0.27 | 0.6064 | 0.77 | 0.7794 | 1.27 | 0.8980 | 1.77 | 0.9616 | 2.27 | 0.98840 | 2.77 | 0.99720 |
| 0.28 | 0.6103 | 0.78 | 0.7823 | 1.28 | 0.8997 | 1.78 | 0.9625 | 2.28 | 0.98870 | 2.78 | 0.99728 |
| 0.29 | 0.6141 | 0.79 | 0.7852 | 1.29 | 0.9015 | 1.79 | 0.9633 | 2.29 | 0.98899 | 2.79 | 0.99736 |
| 0.30 | 0.6179 | 0.80 | 0.7881 | 1.30 | 0.9032 | 1.80 | 0.9641 | 2.30 | 0.98928 | 2.80 | 0.99744 |
| 0.31 | 0.6217 | 0.81 | 0.7910 | 1.31 | 0.9049 | 1.81 | 0.9649 | 2.31 | 0.98956 | 2.81 | 0.99752 |
| 0.32 | 0.6255 | 0.82 | 0.7939 | 1.32 | 0.9066 | 1.82 | 0.9656 | 2.32 | 0.98983 | 2.82 | 0.99760 |
| 0.33 | 0.6293 | 0.83 | 0.7967 | 1.33 | 0.9082 | 1.83 | 0.9664 | 2.33 | 0.99010 | 2.83 | 0.99767 |
| 0.34 | 0.6331 | 0.84 | 0.7995 | 1.34 | 0.9099 | 1.84 | 0.9671 | 2.34 | 0.99036 | 2.84 | 0.99774 |
| 0.35 | 0.6368 | 0.85 | 0.8023 | 1.35 | 0.9115 | 1.85 | 0.9678 | 2.35 | 0.99061 | 2.85 | 0.99781 |
| 0.36 | 0.6406 | 0.86 | 0.8051 | 1.36 | 0.9131 | 1.86 | 0.9686 | 2.36 | 0.99086 | 2.86 | 0.99788 |
| 0.37 | 0.6443 | 0.87 | 0.8078 | 1.37 | 0.9147 | 1.87 | 0.9693 | 2.37 | 0.99111 | 2.87 | 0.99795 |
| 0.38 | 0.6480 | 0.88 | 0.8106 | 1.38 | 0.9162 | 1.88 | 0.9699 | 2.38 | 0.99134 | 2.88 | 0.99801 |
| 0.39 | 0.6517 | 0.89 | 0.8133 | 1.39 | 0.9177 | 1.89 | 0.9706 | 2.39 | 0.99158 | 2.89 | 0.99807 |
| 0.40 | 0.6554 | 0.90 | 0.8159 | 1.40 | 0.9192 | 1.90 | 0.9713 | 2.40 | 0.99180 | 2.90 | 0.99813 |
| 0.41 | 0.6591 | 0.91 | 0.8186 | 1.41 | 0.9207 | 1.91 | 0.9719 | 2.41 | 0.99202 | 2.91 | 0.99819 |
| 0.42 | 0.6628 | 0.92 | 0.8212 | 1.42 | 0.9222 | 1.92 | 0.9726 | 2.42 | 0.99224 | 2.92 | 0.99825 |
| 0.43 | 0.6664 | 0.93 | 0.8238 | 1.43 | 0.9236 | 1.93 | 0.9732 | 2.43 | 0.99245 | 2.93 | 0.99831 |
| 0.44 | 0.6700 | 0.94 | 0.8264 | 1.44 | 0.9251 | 1.94 | 0.9738 | 2.44 | 0.99266 | 2.94 | 0.99836 |
| 0.45 | 0.6736 | 0.95 | 0.8289 | 1.45 | 0.9265 | 1.95 | 0.9744 | 2.45 | 0.99286 | 2.95 | 0.99841 |
| 0.46 | 0.6772 | 0.96 | 0.8315 | 1.46 | 0.9279 | 1.96 | 0.9750 | 2.46 | 0.99305 | 2.96 | 0.99846 |
| 0.47 | 0.6808 | 0.97 | 0.8340 | 1.47 | 0.9292 | 1.97 | 0.9756 | 2.47 | 0.99324 | 2.97 | 0.99851 |
| 0.48 | 0.6844 | 0.98 | 0.8365 | 1.48 | 0.9306 | 1.98 | 0.9761 | 2.48 | 0.99343 | 2.98 | 0.99856 |
| 0.49 | 0.6879 | 0.99 | 0.8389 | 1.49 | 0.9319 | 1.99 | 0.9767 | 2.49 | 0.99361 | 2.99 | 0.99861 |

| $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.00 | 0.5000 | 0.50 | 0.6915 | 1.00 | 0.8413 | 1.50 | 0.9332 | 2.00 | 0.97725 | 2.50 | 0.99379 |
| 0.01 | 0.5040 | 0.51 | 0.6950 | 1.01 | 0.8438 | 1.51 | 0.9345 | 2.01 | 0.97778 | 2.51 | 0.99396 |
| 0.02 | 0.5080 | 0.52 | 0.6985 | 1.02 | 0.8461 | 1.52 | 0.9357 | 2.02 | 0.97831 | 2.52 | 0.99413 |
| 0.03 | 0.5120 | 0.53 | 0.7019 | 1.03 | 0.8485 | 1.53 | 0.9370 | 2.03 | 0.97882 | 2.53 | 0.99430 |
| 0.04 | 0.5160 | 0.54 | 0.7054 | 1.04 | 0.8508 | 1.54 | 0.9382 | 2.04 | 0.97932 | 2.54 | 0.99446 |
| 0.05 | 0.5199 | 0.55 | 0.7088 | 1.05 | 0.8531 | 1.55 | 0.9394 | 2.05 | 0.97982 | 2.55 | 0.99461 |

| z | 0.00 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.06 | 0.07 | 0.08 | 0.09 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | 0.500000 | 0.503989 | 0.507978 | 0.511967 | 0.515953 | 0.519939 | 0.532922 | 0.527903 | 0.531881 | 0.535856 |
| 0.1 | 0.539828 | 0.543795 | 0.547758 | 0.551717 | 0.555760 | 0.559618 | 0.563559 | 0.567495 | 0.571424 | 0.575345 |
| 0.2 | 0.579260 | 0.583166 | 0.587064 | 0.590954 | 0.594835 | 0.598706 | 0.602568 | 0.606420 | 0.610261 | 0.614092 |
| 0.3 | 0.617911 | 0.621719 | 0.625516 | 0.629300 | 0.633072 | 0.636831 | 0.640576 | 0.644309 | 0.648027 | 0.651732 |
| 0.4 | 0.655422 | 0.659097 | 0.662757 | 0.666402 | 0.670031 | 0.673645 | 0.677242 | 0.680822 | 0.684386 | 0.687933 |
| 0.5 | 0.691462 | 0.694974 | 0.698468 | 0.701944 | 0.705401 | 0.708840 | 0.712260 | 0.715661 | 0.719043 | 0.722405 |

| $z$ | 0.00 | 0.01 | 0.02 | 0.03 |
|---|---|---|---|---|
| 0 | 0.50000 | 0.50399 | 0.50800 | 0.51197 |
| $\vdots$ | | $\vdots$ | | |
| 1.5 | 0.93319 | 0.93448 | 0.93574 | 0.93699 |

Example: Assume that a $Z$ is standard normal random variable. Find $P[Z \leq 1.5]$.




| $z$ | 0.00 | 0.01 | 0.02 | 0.03 |
|---|---|---|---|---|
| 0 | 0.50000 | 0.50399 | 0.50800 | 0.51197 |
| ⋮ | | ⋮ | | |
| 1.5 | 0.93319 | 0.93448 | 0.93574 | 0.93699 |

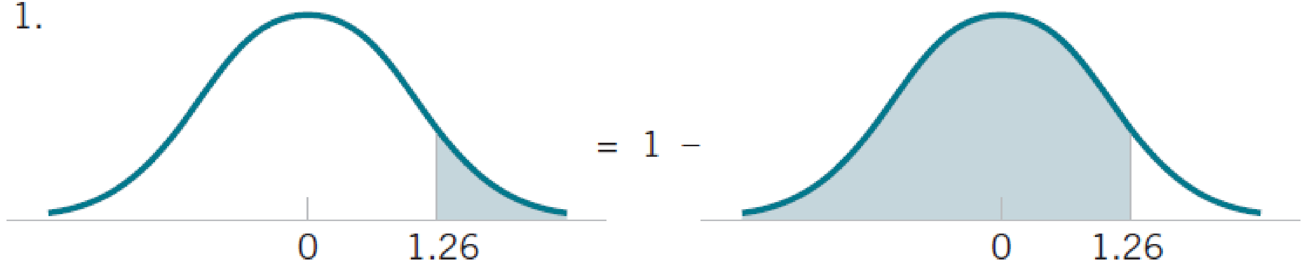
| $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ | $z$ | $\Phi(z)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.00 | 0.5000 | 0.50 | 0.6915 | 1.00 | 0.8413 | 1.50 | 0.9332 | 2.00 | 0.97725 | 2.50 | 0.99379 |

Example: <u>Following probabilities can also be found from the table</u>.

1. $P[Z > 1.26]$
2. $P[\ Z < \ -1.26]$
3. $P[Z > \ -1.26]$
4. $P[-1.26 < Z < 1.26]$
5. $P[Z \leq 4.6]$
6. Find the value of $z$ such that $P[Z > z] = 0.05$
7. Find the value of $z$ such that $P[-z < Z < z] = 0.99$

1. $P[Z > 1.26]$

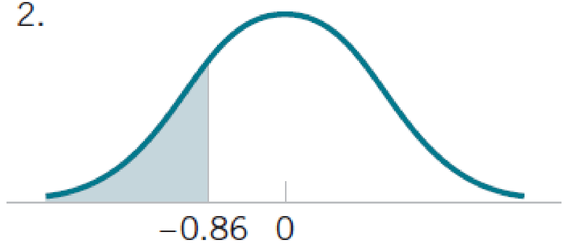
1.



2. $P[\,Z < -1.26]$

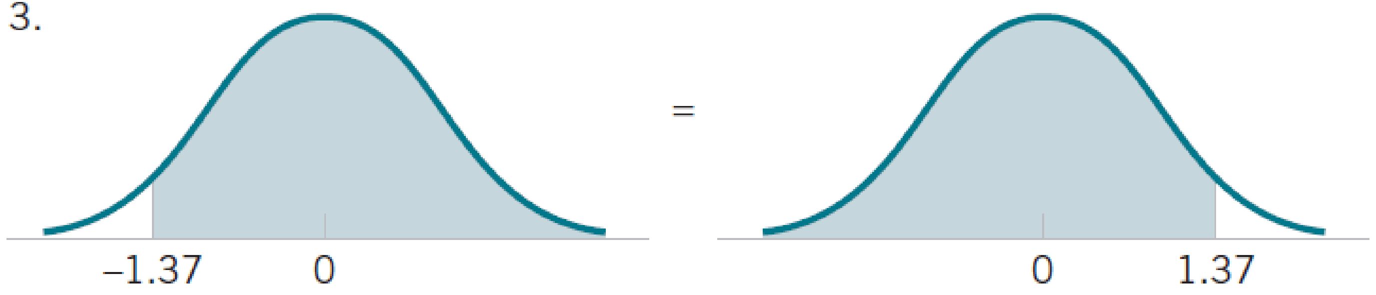
$$P[Z < -1.26] = P[Z > 1.26]$$

2.

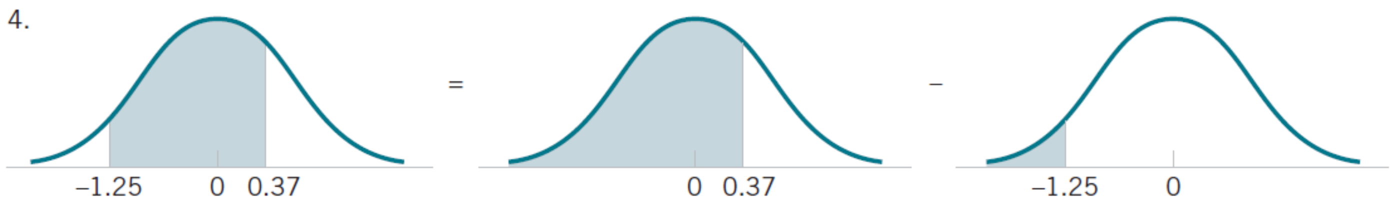


3. $P[Z > -1.26]$

3.

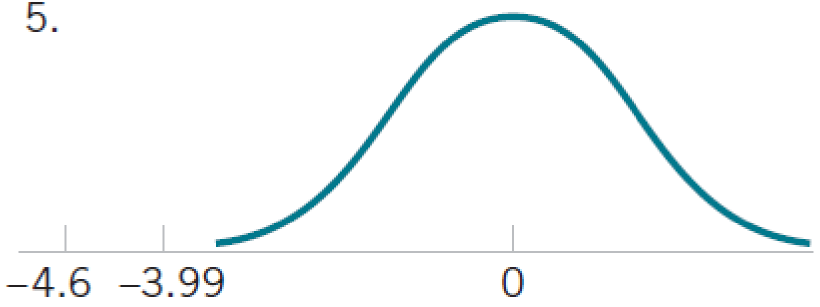


4. $P[-1.26 < Z < 1.26]$



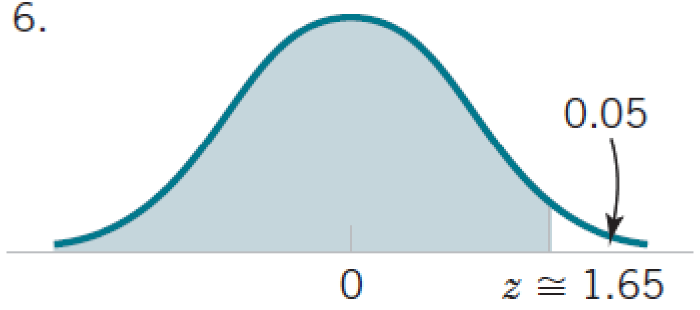
5. $P[Z \leq 4.6]$

5.



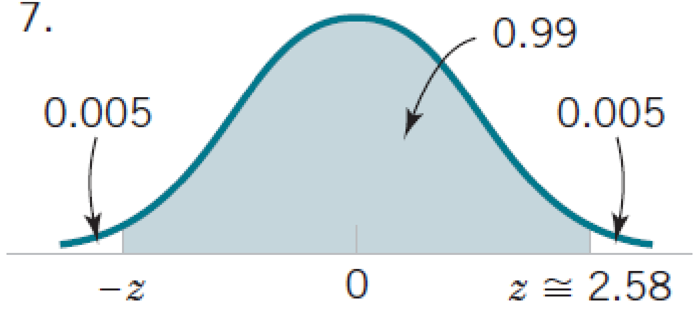
6. Find the value of $z$ such that $P[Z > z]$

6.



7. Find the value of $z$ such that $P[-z < Z < z] = 0.99$

7.



Standardizing a Normal Random Variable

If $X$ is a normal random variable with $E(X) = \mu$ and $V(X) = \sigma^2$, the random variable

$$Z = \frac{X - \mu}{\sigma}$$

is a normal random variable with $E(Z) = 0$ and $V(Z) = 1$. That is, $Z$ is a standard random variable.

Creating a new random variable by this transformation is referred to as **standardizing** or transforming normal random variable into its standard form is referred to as standardization.

How do we believe that Z has the zero mean and unit variance?

Example: Suppose that the current measurements in a strip of wire are assumed to follow a normal distribution with a mean of 10 milliamperes and a variance of 4 (milliamperes)$^2$. What is the probability that a measurement exceeds 13 milliamperes?

Solution:

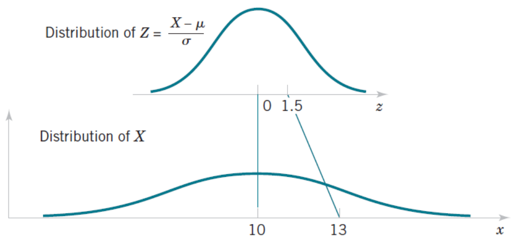
Let $X$ denote the current in milliamperes $\quad P[X > 13] = ?$

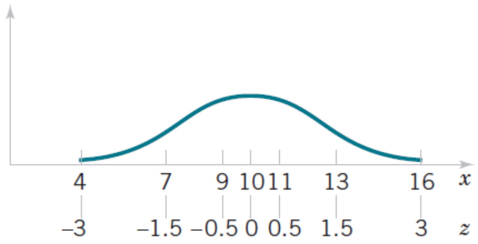
And $Z$ denote a standard normal RV $\quad Z = \frac{X-\mu}{\sigma}$

$$Z = \frac{X - 10}{2}$$

Note that $X > 13$ corresponds to $Z > 1.5$

$$P(X > 13) = P(Z > 1.5) = 1 - P(Z \leq 1.5)$$
$$= 1 - 0.93319 = 0.06681$$



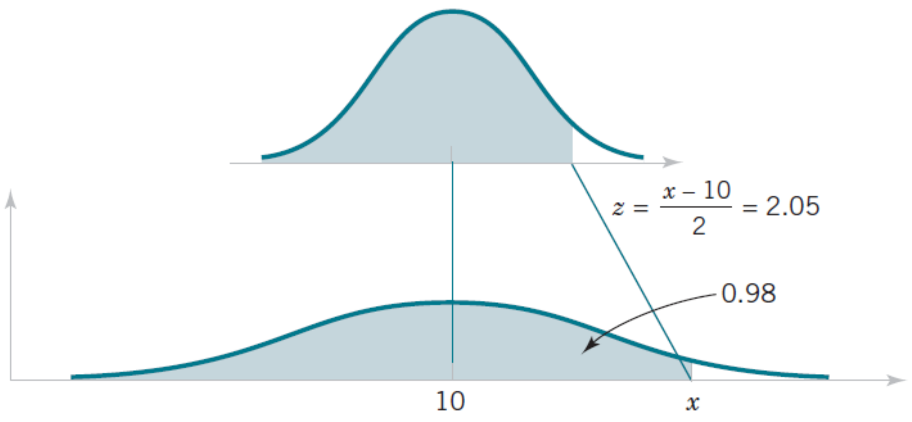
### Standardizing to Calculate a Probability

Suppose that $X$ is a normal random variable with mean $\mu$ and variance $\sigma^2$. Then,

$$P(X \leq x) = P\left(\frac{X-\mu}{\sigma} \leq \frac{x-\mu}{\sigma}\right) = P(Z \leq z) \tag{4.11}$$

where $Z$ is a **standard normal random variable**, and $z = \dfrac{(x-\mu)}{\sigma}$ is the **$z$-value** obtained by **standardizing** $X$. The probability is obtained by using Appendix Table III with $z = (x-\mu)/\sigma$.

Example (Continuing with the previous example): Suppose that the current measurements in a strip of wire are assumed to follow a normal distribution with a mean of 10 milliamperes and a variance of 4 (milliamperes)$^2$. What is the probability that a current measurement is between 9 and 11 milliamperes?



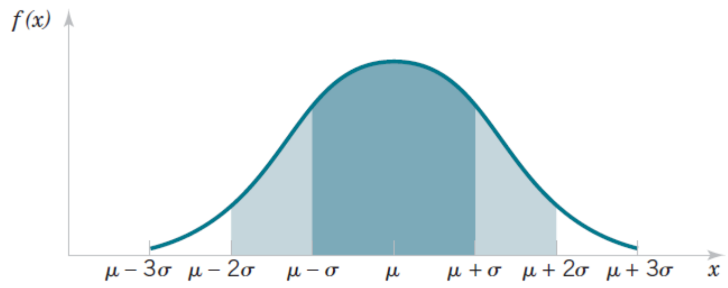
Solution:

$$
\begin{aligned}
P(9 < X < 11) &= P\left(\frac{9-10}{2} < \frac{X-10}{2} < \frac{11-10}{2}\right) \\
&= P(-0.5 < Z < 0.5) \\
&= P(Z < 0.5) - P(Z < -0.5) \\
&= 0.69146 - 0.30854 = 0.38292
\end{aligned}
$$

Example: Let $X$, the grade of a randomly selected student in a test of a probability

course, be a normal random variable. A professor is said to grade such a test *on the curve* if he finds the average $\mu$ and the standard deviation $\sigma$ of the grades and then assigns letter grades according to the following table.

| Range of the grade | $X \geq \mu+\sigma$ | $\mu \leq X < \mu+\sigma$ | $\mu-\sigma \leq X < \mu$ | $\mu-2\sigma \leq X < \mu-\sigma$ | $X < \mu-2\sigma$ |
|---|---|---|---|---|---|
| Letter grade | A | B | C | D | F |



$$P(X \geq \mu + \sigma) = P\Big(\frac{X - \mu}{\sigma} \geq 1\Big) = 1 - \Phi(1) \approx 0.1587.$$

$$P(\mu \leq X < \mu + \sigma) = P\Big(0 \leq \frac{X - \mu}{\sigma} < 1\Big) = \Phi(1) - \Phi(0) \approx 0.3413$$

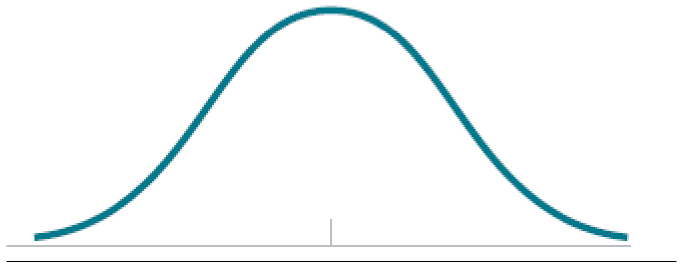
$$
\begin{aligned}
P(\mu-\sigma \le X < \mu) &= P\Big(-1 \le \frac{X-\mu}{\sigma} < 0\Big) = \Phi(0) - \Phi(-1) \\
&= 0.5 - 0.1587 \approx 0.3413, \\
P(\mu-2\sigma \le X < \mu-\sigma) &= P\Big(-2 \le \frac{X-\mu}{\sigma} < -1\Big) = \Phi(-1) - \Phi(-2) \\
&= 0.1587 - 0.0228 \approx 0.1359, \\
P(X < \mu-2\sigma) &= P\Big(\frac{X-\mu}{\sigma} < -2\Big) = \Phi(-2) \approx 0.0228.
\end{aligned}
$$

Example: The scores on an achievement test given to the students at a university are normally distributed with mean 500 and standard deviation 100. What should the score of a student be to place him among the top 10% of all students?

Solution:

Let X be a normal RV with mean 500 and SD 100

We need to find a $x$, so that $P[X \geq x] = 0.10$ or $P[X < x] = 0.90$



$$P\Big(\frac{X-500}{100} < \frac{x-500}{100}\Big) = 0.90.$$

$$\Phi\Big(\frac{x-500}{100}\Big) = 0.90.$$

We need to apply the inverse CDF function to calculate the value of $x$.

From the table, we have $\Phi(1.28) \approx 0.8997$

$$\frac{x-500}{100} \approx 1.28 => x \approx 628$$

Therefore, a student should earn 628 or more to be among the top $10\%$ of the students.

Linear Combination of Normally distributed RVs

If the random variables $X_1, \ldots, X_k$ are independent and if $X_i$ has the normal distribution with mean $\mu_i$ and variance $\sigma^2$, $(i = 1, \ldots, k)$, then the sum

$X_1 + \ldots + X_k$

has the normal distribution with mean

$$\mu_1 + \cdots + \mu_k$$

and variance

$$\sigma_1^2 + \cdots + \sigma_k^2$$

If the random variables $X_1, \ldots, X_k$ are independent, if $X_i$ has the normal distribution with mean $\mu_i$ and variance $\sigma_i^2$ $(i = 1, \ldots, k)$, and if $a_1, \ldots, a_k$ and $b$ are constants for which at least one of the values $a_1, \ldots, a_k$ is different from 0, then the variable $a_1X_1 + \cdots + a_kX_k + b$ has the normal distribution with mean $a_1\mu_1 + \cdots + a_k\mu_k + b$ and variance $a_1^2\sigma_1^2 + \cdots + a_k^2\sigma_k^2$. ■

Example:

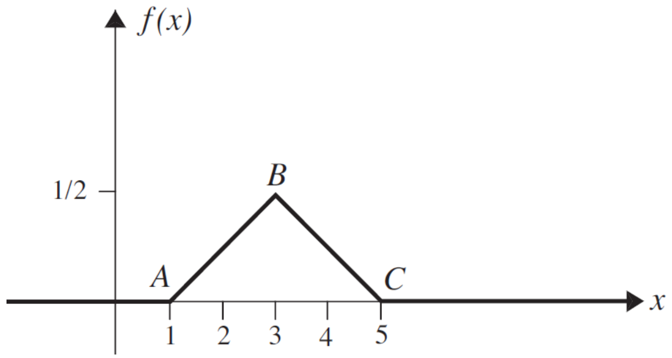
**Heights of Men and Women.** Suppose that the heights, in inches, of the women in a certain population follow the normal distribution with mean 65 and standard deviation 1, and that the heights of the men follow the normal distribution with mean 68 and standard deviation 3. Suppose also that one woman is selected at random and, independently, one man is selected at random. We shall determine the probability that the woman will be taller than the man.

Lognormal Distribution:

If $log(X)$ has the normal distribution with mean $\mu$ and variance $\sigma^2$, we say that $X$ has the *lognormal distribution* with parameters $\mu$ and $\sigma^2$.

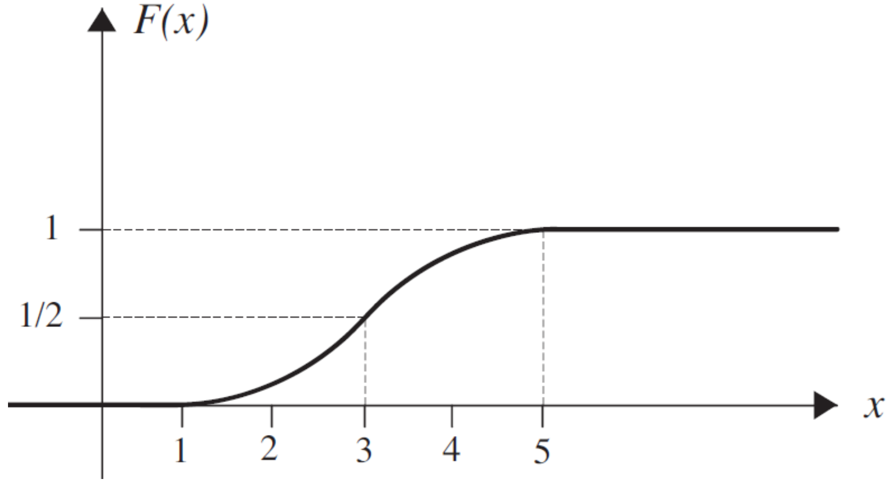
Triangular Distribution: The PDF of the RV $X$ has the following PDF:

$$f(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{2} + \frac{1}{4}(x-3) & 1 \le x < 3 \\ \frac{1}{2} - \frac{1}{4}(x-3) & 3 \le x < 5 \\ 0 & x \ge 5. \end{cases}$$



The CDF of $X$ is given by

$$F(t) = \begin{cases} 0 & t < 1 \\ \frac{1}{8}t^2 - \frac{1}{4}t + \frac{1}{8} & 1 \leq t < 3 \\ -\frac{1}{8}t^2 + \frac{5}{4}t - \frac{17}{8} & 3 \leq t < 5 \\ 1 & t \geq 5. \end{cases}$$


F(x)
1
1/2
x
1
2
3
4
5

## Laplace Distribution

**Laplace Distributions.** Let $\sigma > 0$ and $\theta$ be real numbers. The distribution whose p.d.f. is

$$f(x|\theta, \sigma) = \frac{1}{2\sigma} e^{-|x-\theta|/\sigma} \tag{10.7.5}$$

is called the *Laplace distribution with parameters $\theta$ and $\sigma$*.

The random variable $X$ is said to be a **Laplace random variable** or **double exponentially distributed** if its probability density function is given by

$$f(x) = ce^{-|x|}, \quad -\infty < x < +\infty.$$